Algorithmic Bias in Predictive Clinical Decision Support: An Ethical Evaluation

The application of clinical decision-support systems (CDSS) and predictive models in biomedical informatics raises several ethical concerns, including algorithmic bias and the potential for unjust resource allocation. Since predictive models are often trained on EHRs and commercial claims, they may reflect historical trends of social inequality rather than promote health equity. For instance, Obermeyer et al. (2019) found that commercial risk-stratification algorithms that determine access to care for chronic disease management systems scored black patients lower than white ones with similar needs. Such scoring resulted from the algorithm's design, which utilized the amount of healthcare spending as an indicator of morbidity. In other words, because of social inequities in healthcare access, black patients were labeled as healthier than they were due to their spending patterns. Hence, the number of black patients enrolled in care-management programs was significantly lower than that of white patients. The ethical concerns that biomedical informaticians should consider when implementing such algorithms include utilitarianism and deontology.

Utilitarianism, as a consequentialist moral philosophy, was developed by Jeremy Bentham and John Stuart Mill. The ethical theory suggests that the morality of an action is determined by its ability to maximize well-being (good) and minimize evil (bad). In health informatics, utilitarianism is often applied to justify the use of various predictive models that seek to improve the functioning of institutions while minimizing expenses. From the perspective of this ethical theory, the implementation of an automated triage system is justified by the fact that it should benefit the majority of patients by ensuring the most efficient use of resources. However, a utilitarian assessment of the proposed decision would be compromised by the use of an algorithm that works on incorrect assumptions. The fact that such a system would deny access to those needing chronic disease management will lead to increased morbidity and, as a consequence, to even higher costs for the system due to readmissions. One of the weaknesses of utilitarianism is the potential for the majority to receive benefits at the expense of the well-being of the few. Thus, the use of a flawed algorithm

would violate the principle of non-maleficence and undermine the trust of the community in the digital health system, which cannot be justified by its aggregate benefits.

Developed by Immanuel Kant, **Deontology** focuses on actions being morally right or wrong according to fixed duties and rational principles, independent of their outcomes. According to the categorical imperative, moral agents must never treat another human being as a means but rather as an end. Using the cost-benefit index as a measure of patient suffering in predictive informatics would go against the categorical imperative by viewing people as objects to be used for the sake of a larger organization's prosperity. Secondly, a health organization cannot universally will the maxim of using predictive diagnosis software that delivers different accuracy rates amongst demographic groups and puts patients at risk. This practice would contradict the duty of care and distributive justice moral frameworks. Deontology requires informaticians to have a moral obligation to ensure software used in healthcare diagnosis is free from bias and meets standards of equality before being deployed in clinical settings, regardless of organizational benefits.

Deployment of unaudited predictive risk tools violates both outcome- and duty-based ethical frameworks. This evidence supports the need for informaticians to conduct a comprehensive audit of bias prior to deployment and ongoing algorithmic monitoring to avoid technological health disparities. The AHIMA code of ethics mandates that health information technology be used in a manner that upholds the dignity of all individuals in its use.

References

- McWay, D. C. (2021). Legal and ethical aspects of health information management (Fifth ed.). Cengage Learning.
- American Health Information Management Association. (2019). *AHIMA Code of Ethics*. AHIMA House of Delegates.
- Kant, I. (1785). *Groundwork of the metaphysics of morals* (M. Gregor, Trans.). Cambridge University Press.

- Mill, J. S. (1863). *Utilitarianism*. Parker, Son, and Bourn.
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447–453. https://doi.org/10.1126/science.aax2342